

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
/* CONSTANT TIME OPERATION  $O(1)$  */
```

```
void constantTime(int n) {
```

```
    int a = 10, b = 20, c;
```

```
    c = a + b; // Single operation
```

```
}
```

```
/* LINEAR TIME OPERATION  $O(n)$  */
```

```
void linearTime(int n) {
```

```
    int *arr = (int*)malloc(n * sizeof(int)); // Extra space  $O(n)$ 
```

```
    int i;
```

```
    for(i = 0; i < n; i++) {
```

```
        arr[i] = i;
```

```
    }
```

```
    free(arr);
```

```
}
```

```
/* QUADRATIC TIME OPERATION  $O(n^2)$  */
```

```
void quadraticTime(int n) {
```

```
    int **mat = (int**)malloc(n * sizeof(int*)); // Space  $O(n^2)$ 
```

```
    int i, j;
```

```
    for(i = 0; i < n; i++) {
```

```
        mat[i] = (int*)malloc(n * sizeof(int));
```

```
    }
```

```
    for(i = 0; i < n; i++) {
```

```
        for(j = 0; j < n; j++) {
```

```
            mat[i][j] = i + j;
```

```
    }  
}  
  
for(i = 0; i < n; i++) {  
    free(mat[i]);  
}  
free(mat);  
}  
  
int main() {  
    int n;  
  
    printf("Enter input size: ");  
    scanf("%d", &n);  
  
    constantTime(n);  
    linearTime(n);  
    quadraticTime(n);  
  
    printf("Operations completed for n = %d\n", n);  
  
    return 0;  
}
```